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Device for leveling and aligning tiles and method for leveling and aligning tiles

Abstract

A device for leveling and aligning tiles and method for leveling and aligning tiles are disclosed. In one embodiment, the leveling device includes a body and two spaced and parallel strip members extending transversely from the body. Each of the spaced and parallel strip members extend to the front and rear of the body. Two opposing lateral open windows are formed in the body. A breakaway section is defined along the body. A wedge device is provided for penetrating one or more of the two opposing lateral open windows and exerting a force on the tiles for leveling them relative to each other.

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Background/Summary

PRIORITY STATEMENT & CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of co-pending U.S. patent application Ser. No. 17/516,095, entitled “Device for Leveling and Aligning Tiles and Method for Leveling and Aligning Tiles” filed Nov. 1,

2021, in the names of Clinton D. Bunch et al.; which is a continuation of U.S. patent application Ser. No. 16/685,036, entitled "Device for Leveling and Aligning Tiles and Method for Leveling and Aligning Tiles" filed Nov. 15, 2019, in the names of Clinton D. Bunch et al., and issued on Nov. 2, 2021 as U.S. Pat. No. 11,162,266; which is a continuation of U.S. patent application Ser. No. 16/102,344, entitled "Device for Leveling and Aligning Tiles and Method for Leveling and Aligning Tiles" filed on Aug. 13, 2018, in the names of Clinton D. Bunch et al., and issued on Jul. 7, 2020 as U.S. Pat. No. 10,704,274; which is a continuation of U.S. patent application Ser. No. 15/345,802, entitled "Device for Leveling and Aligning Tiles and Method for Leveling and Aligning Tiles" filed on Nov. 8, 2016, in the names of Clinton D. Bunch et al., and issued on Aug. 14, 2018 as U.S. Pat. No. 10,047,530; which is a continuation of U.S. patent application Ser. No. 15/044,907 entitled "Device for Leveling and Aligning Tiles and Method for Leveling and Aligning Tiles" filed on Feb. 16, 2016, in the names of Clinton D. Bunch and Joshua A. Bunch, and issued on Nov. 8, 2016 as U.S. Pat. No. 9,487,959; which is a continuation-in-part of U.S. patent application Ser. No. 13/859,316 entitled "Device for Leveling and Aligning Tile and Method for Leveling and Aligning Tiles" filed on Apr. 9, 2013, in the names of Clinton D. Bunch and Joshua A. Bunch, and issued on Feb. 16, 2016 as U.S. Pat. No. 9,260,872; all of which are hereby incorporated by reference, in entirety, for all purposes.

TECHNICAL FIELD OF THE INVENTION

(1) This invention relates, in general, to tile installation and, in particular to a device for leveling and aligning tiles and properly spacing tiles during the installation thereof.

BACKGROUND OF THE INVENTION

(2) Tile has become a popular decorative and functional article for use in floors, walls, countertops, and the like. Both professional tile installers and do-it-yourselfers spend a great deal of time aligning and leveling tiles as they are being placed on a substrate's surface. Proper alignment and leveling of each tile is important for a number of reasons. Improper installation can cause the need for tiles to be replaced in order to prevent a spacing error from propagating across the substrate, aesthetic reasons, and in some instances, safety concerns. A need exists for a device for leveling and aligning tiles and properly spacing tiles.

SUMMARY OF THE INVENTION

(3) It would be advantageous to achieve a device for leveling and aligning tiles and properly spacing tiles. It would also be desirable to enable a mechanical-based solution that furnishes an inexpensive tool that assists professional tile installers and do-it-yourselfers. To better address one or more of these concerns, in one aspect of the invention, a tile leveling device and a wedge device for use with tiles are disclosed. In one embodiment, the leveling device includes a body and two spaced and parallel strip members extending transversely from the body. Each of the spaced and parallel strip members extend to the front and rear of the body. Two opposing lateral open windows are formed in the body. A breakaway section is defined along the body.

(4) In one implementation, a wedge device includes a backstop member and a pair of members extending from the backstop member. Each of pair of members includes a tapered surface configured to penetrate the lateral open windows and exerting force against tiles, thereby pressing the tiles against the first and second strip members in order to level and align.

(5) In another embodiment, the leveling device includes an inverted U-shaped body with an I-shaped base orthogonally coupled thereto. An open window is located within the U-shaped body between stems thereof. The I-shaped base has four bars extending transversely from the inverted U-shaped body and that extend upwards toward the inverted U-shaped body in an arcuate fashion to define respective four points of contact for two, three, and four tiles. Two breakaway sections are respectively defined along the spaced stems of the inverted U-shaped body at the I-shaped bar. In another implementation, a wedge device includes an oversized backstop member and a wedge

member extending from the oversized backstop member. These and other aspects of the invention will be apparent from and elucidated with reference to the embodiments described hereinafter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) For a more complete understanding of the features and advantages of the present invention, reference is now made to the detailed description of the invention along with the accompanying figures in which corresponding numerals in the different figures refer to corresponding parts and in which:
- (2) FIGS. **1A** and **1B** are front perspective views of one embodiment of a leveling device with a wedge device according to the teachings presented herein;
- (3) FIG. **2** is a side view, in partial cross-section, of the leveling device with the wedge device presented in FIGS. **1A** and **1B**;
- (4) FIGS. **3** through **5** are top plan views showing installations of two, three, and four tile using the leveling device and wedge device presented in FIGS. **1A**, **1B**, and **2**;
- (5) FIGS. **6** through **8** are side cross-sectional views of one embodiment of the installation of tile on a floor using the leveling device and wedge device presented in FIGS. **1A**, **1B**, and **2**;
- (6) FIG. **9** is a front cross-sectional view of the embodiment of the installation of tile on a floor or subsurface using the leveling device and wedge device presented in FIGS. **1A**, **1B**, and **2**;
- (7) FIG. **10** is a front perspective view of another embodiment of a leveling device according to the teachings presented herein;
- (8) FIGS. **11A** and **11B** are front perspective views of one embodiment of a leveling device with a wedge device according to the teachings presented herein;
- (9) FIG. **12** is a side view, in partial cross-section, of the leveling device with the wedge device presented in FIGS. **11A** and **11B**;
- (10) FIG. **13** is a front perspective view of an alternate embodiment of a leveling device;
- (11) FIG. **14** is a side elevation view of the alternative embodiment of the leveling device depicted in FIG. **13**.
- (12) FIG. **15** is a front perspective view of a further alternate embodiment of a leveling device; and
- (13) FIG. **16** is a bottom plan view of the leveling device depicted in FIG. **15**.

DETAILED DESCRIPTION OF THE INVENTION

(14) While the making and using of various embodiments of the present invention are discussed in detail below, it should be appreciated that the present invention provides many applicable inventive concepts which can be embodied in a wide variety of specific contexts. The specific embodiments discussed herein are merely illustrative of specific ways to make and use the invention, and do not delimit the scope of the present invention.

(15) Referring initially to FIGS. **1A**, **1B**, and **2**, therein is depicted one embodiment of a tile leveling device that is schematically illustrated and generally designated **10**. The tile leveling device **10** and a wedge device **12** are utilized, in combination, to align and level two, three, or four tiles, for example. The leveling device **10** includes a body **14** and spaced and parallel strip members **16**, **18** extending transversely from the body **14**. Each of the spaced and parallel strip members **16**, **18** extend to the front F and rear R of the body **14**. Lateral open windows **20**, **22** having upper edges **21**, **23** are formed in the body **14** and sized to accept a member having a tapered surface configured to penetrate the respective lateral open windows **20**, **22** and exert force thereunder. A breakaway section **24** is defined along the body **14**. The breakaway section **24** may be a frangible section of the body **14** of reduced thickness that would promote the breakaway, and thus, separation of the body **14**. An upper bump **26** and a lower bump **28** extend horizontally across the body **14**. The upper bump **26** being larger and more pronounced to provide lifting power. A spacing pad **30** is

integral with the body **14** and may vary in thickness depending on the application. The spacing pad **30** contributes to furnishing a combination of vertical leveling and joint spacing within a single product. Moreover, the spacing pad **30**, which may be more generally a spacer, is configured to position the tiles a predetermined distance apart depending on the application.

(16) The spaced and parallel strip members **16, 18** provide four points of contact **32, 34, 36, 38** for lift of tiles, while still establishing space for maximum mortar penetration between the spaced and parallel strip members **16, 18**. As shown, the spaced and parallel strip members **16, 18** have an arcuate form wherein the spaced and parallel strip members **16, 18** curve upwards from the junction with the body **14** to the four points of contact **32, 34, 36, 38**. That is, the spaced and parallel strip members **16, 18** define convex curvatures **40, 42** that are flexible and compressible for tile installation during a leveling and alignment of a tile. The convex curvatures **40, 42** also ensure that tiles of varying thicknesses may be leveled and aligned. In fact, in the corner embodiment, four tiles having four varying thicknesses may be leveled and aligned by way of the flexibility and compressibility of the convex curvatures **40, 42** of the strip members **16, 18**.

(17) The wedge device **12** includes a backstop member **50** and two extension members, depicted as wedge members **52, 54** extending from the backstop member **50**. The backstop member provides a push area for fingers or a thumb and an enhanced sized that furnishes more leverage during use. Each of the wedge members **52, 54** include respective tapered surfaces **56, 58** configured to penetrate the lateral open windows **20, 22** and exert force against the tiles by pressing the tiles against the strip members **16, 18**. Teeth **60, 62** are located along the tapered surfaces **56, 58** in order to latch onto the respective upper edges **21, 23** of the opposing lateral open windows **20, 22**. In operation, the teeth **60, 62** prevent the respective wedge members **52, 54** from slipping out of the lateral open windows **20, 22** during penetration thereof. As will be appreciated, the wedge device **12** may penetrate the leveling device from the front F or rear R.

(18) Referring now to FIGS. 3 through 5, the tile leveling device may be utilized with two tiles **70, 72** (FIG. 3), three tiles (FIG. 4) **74, 76, 78**, or four tiles **80, 82, 84, 86** (FIG. 5) for installation on a substrate, subsurface, or other surface, which is indicated by the letter S. By way of example, in the two tile installation, the tile **70** is positioned over the front portions of the strip members **16, 18**. The tile **70** has a lower surface **88** opposite an upper surface **90**, wherein the lower surface **88** faces the strip members **16, 18** and the subsurface S. The upper surface **90** is farther from the strip members **16, 18** than the lower surface **88** and faces away from the strip members **16, 18**. The second tile **72** is similarly situated over the rear portions of the strip members **16, 18** and includes a lower surface **92** and an upper surface **94**.

(19) Accordingly, a single leveling device **10** and wedge device **12** may be utilized to install, align, and level between two and four tiles. The use of the wedge device **12** having two extension members and the two lateral open windows **20, 22** provide for utilization at corner tiles; thereby offering improved efficiency. More particularly, using one leveling device for a corner improves efficiency and minimizes the number of leveling devices required to complete a job. Additionally, as previously discussed, the arcuate portions of the parallel strip members **16, 18** compress and flatten to accommodate different thicknesses of tiles to provide a level surface. In fact, the leveling device and wedge device presented herein may simultaneously accommodate between two and four different thicknesses of tiles.

(20) Referring now to FIGS. 6 through 9, one example of installation is shown wherein the tile leveling device **10** is placed such that one end of the tile **72** sits on the rear portions of the parallel strip members **16, 18** and the tile **72** abuts the body **14** of the leveling device. Such an installation methodology may be used for any of the configurations shown in FIGS. 3 through 5, for example. By way of illustration, therefore, this example is for the two tile application of FIG. 3. The adjacent tile **70** sits on the other side of the body **14** and across the front portions of the parallel strip members **16, 18**. The extension members, namely wedge members **52, 54** of the wedge device **12** are then inserted in the respective lateral open windows **20, 22**. The backstop member **50** is then

pushed, thereby driving the wedge members **52, 54** to penetrate the respective lateral open windows **20, 22**.

(21) In one embodiment, the wedge device **12** offers an advantage over individual wedges as the backstop member **50** provides a wider or broader surface to push. In particular, as shown in FIG. **8**, a thumb or finger may be used to drive the wedge device **12** into the leveling device **10**. The ease of push advantage translates into a more powerful lift because of the more even distribution of the lifting surface. Therefore, the leveling device **10** and wedge device **12** combination provide improved performance, both quantitatively and qualitatively. With respect to the former, the lateral open windows **20, 22** in combination with the wedge device **12** permit use at corners increasing efficiency and reducing the number of leveling and aligning operations, which leads to faster installation using fewer leveling devices. With respect to the latter, the improved contact surface provided by the backstop member **50** minimizes the stress and strain on the fingers and thumbs.

(22) Additionally, the design of the wedge device **12** discourages the tiles from pivoting by providing two points of contact with each tile edge. By way of example, on a 12 inch by 12 inch tile, a traditional installation system would require two spacers for each side, for a total of eight spacers, in order to stabilize the edges and prevent pivoting. Using the teachings presented herein, four leveling devices **10** may be used and the optimal eight points of contact would be maintained.

(23) Continuing with the description of FIGS. **6** through **9**, in one implementation, as these wedge-shaped extension members are continually pushed through the lateral open windows **20, 22** due to the increasing thickness of the wedge members **52, 54**, the action causes the lower surfaces **88, 92** of each of the tiles **70, 72** to be compressed downward pressing the strip members **16, 18** beneath the tiles **70, 72** toward the subsurface S, on which is located mortar M for bonding the tiles **70, 72** to the subsurface S. As a result, the tiles **70, 72** are aligned and leveled. The spacing between the tiles is controlled by the thickness of the body and, in one embodiment, the presence of the spacing pad **30**, which may more generally be a spacer. Once the tiles **70, 72** are set, the wedge device **12** may be removed and the body **14** is broken off by kicking or applying force to the side of the wedge device **12** such that the body **14** is severed at the breakaway section **24**. As shown in FIG. **9**, the application of force by a boot, breaks the body at a break at the breakaway section **24**. In the illustrated implementation, the breakaway section **24** is positioned such that it is located between the surfaces **88, 90, 92, 94** of the tiles **70, 72**. That is, the breakaway section **24** is located at a height within the thickness of the installed tiles. In this position, the breakaway section **24** is not exposed to any adhesive that may be used to adhere the tiles to the subsurface S, which may be a floor or wall, for example.

(24) Referring now to FIG. **10**, an alternative embodiment of the tile leveling device **10** includes the body and the spaced and parallel strip members **16, 18** extending transversely from the body **14**. In this embodiment, opposing lateral closed windows **100, 102** are formed in the body **14** with each of the opposing lateral closed windows **100, 102** sized to accept a wedge device having wedge members including a tapered surface configured to penetrate the respective lateral closed windows **100, 102** and exert force against multiple tiles pressing the tiles against the strip members **16, 18**. Additionally, the breakaway section **24** is defined along the body **14**. By way of further example, a further embodiment is possible, wherein the leveling device **10** would include a lateral open window and a lateral closed window.

(25) Referring to FIGS. **11A, 11B**, and **12**, in one embodiment of a tile leveling device **110** and tile combination with a wedge device **112**, the tile leveling device **110** includes an inverted U-shaped body **114** defining an open window **116** between two stems **118, 120** of the inverted U-shaped body **114**. An I-shaped base **122** is orthogonally coupled to the inverted U-shaped body **114** such that four spaced bars **124, 126, 128, 130** extend transversely from the inverted U-shaped body **114**. In particular, the spaced bars **124, 126** extend to the front F of the inverted U-shaped body **114** and the spaced bars **128, 130** extend to the rear R of the inverted U-shaped body **114**. Each of the bars **124, 126, 128, 130** extends upward toward the inverted U-shaped body in an arcuate fashion to define

respective four points of contact **132, 134, 136, 138** for two, three, and four tiles.

(26) Two breakaway sections **140, 142** are defined along the respective two stems **118, 120** of the inverted U-shaped body **114**. Additionally, as shown, a notch **144** is formed between the bars **124, 126** and a notch **146** is formed between the bars **128, 130**. The open window **116** includes an upper edge **148**. An open span **150** is proximally interposed between the breakaway sections **140, 142** and the open span **150** forms a portion of the open window **116**. Spacing pads, such as spacing pad **152**, may be utilized to position the tiles a predetermined distance apart, depending on the application.

(27) The wedge device **112** includes an oversized backstop member **160**. A wedge member **162** extends from the oversized backstop member **160** and includes a tapered surface **164** that is configured to penetrate the open window **116** and exert force against two, three, or four tiles pressing the tiles against the bars **124, 126, 128, 130**. The breakaway sections **140, 142** are located where the ends of the stems **118, 120** of the inverted U-shaped body **114** contact the I-shaped base **122**. Teeth **166** are positioned along the tapered surface **164** in order to latch onto the upper edge **148** of the open window **116**.

(28) In operation, the leveling device **110** may be used to align two, three or four tiles and operation is similar to leveling device **10** and wedge device **12**, as previously presented. Similar to the leveling device presented in FIG. 5, in a four-tile embodiment, each tile has corner-to-subfloor contact due to the notches that provide space for mortar contact therein. Similarly, in a two-tile implementation, for example, each tile has edge-to-subfloor contact due to the notches. More particularly, the tapered surface **164** penetrates the open window **116** contacting the upper edge **148** thereof and exerting force against both tiles pressing the tiles against the bars **124, 126, 128, 130**, wherein breakaway sections **140, 142** are located beyond the undersurfaces of the tiles in a direction away from the bars **124, 126, 128, 130**. As previously discussed, the arcuate portions of the parallel strip members compress and flatten to accommodate different thicknesses of tiles to provide a level surface. In fact, the leveling device and wedge device presented herein may simultaneously accommodate between two and four different thicknesses of tiles.

(29) Referring now to FIGS. **13** and **14**, a tile leveling device **200** for use with a locking subassembly **202** are presented. As shown, the tile leveling device includes a shaft **204** and spaced and parallel strip members **206, 208** extend transversely from the shaft **204**. The locking subassembly **202** is configured to traverse the shaft and exert force against the tiles by pressing the tiles against the parallel strip members, similar to the functionality described in previous embodiments. Each of the spaced and parallel strip members extend to the front and rear of the shaft **204**. As shown, a frangible breakaway section is defined along the shaft **204**. A spacing pad **207**, which may be similar to spacing pad **30**, may be integral with the shaft **204** and may vary in thickness depending on the application. The spaced and parallel strip members **206, 208** provide four points of contact **210, 212, 214, 216** for lift of tiles, while still establishing space for maximum mortar penetration between the spaced and parallel strip members **206, 208**. Convex curvatures **218, 220** ensure that the tiles of varying thicknesses may be leveled and aligned, including the alignment of up to four tiles of varying thickness.

(30) In operation, once the tiles are properly positioned, the locking subassembly **202** is secured in its place above the tiles and prevented from moving along the shaft **204** before being driven down to compress the tiles. The shaft **204** may include a locking surface **222**, such as a “zip tie” to enable movement along the shaft **204** by the locking subassembly **202** in only one direction, i.e., toward the tiles.

(31) Referring to FIGS. **15** and **16**, in one further embodiment of the tile leveling device **110** and tile combination with the wedge device **112**, the tile leveling device **110** includes the inverted U-shaped body **114** defining the open window **116** between two stems **118, 120** of the inverted U-shaped body **114**. The I-shaped base **122** is orthogonally coupled to the inverted U-shaped body **114** such that four spaced bars **124, 126, 128, 130** extend transversely from the inverted U-shaped body **114**. In particular, the spaced bars **124, 126** extend to the front F of the inverted U-shaped

body **114** and the spaced bars **128, 130** extend to the rear R of the inverted U-shaped body **114**. Each of the bars **124, 126, 128, 130** extends upward toward the inverted U-shaped body in an arcuate fashion to define respective four points of contact **132, 134, 136, 138** for two, three, and four tiles.

(32) As shown, the I-shaped base is intersected by a crossbar **230**, which is located between the bars **124, 126** and the bars **128, 130**. The crossbar **230** may take any shape or form and may be considered a bi-directional projection, for example. As illustrated, the crossbar **230** includes a bar **232** extending to the front F of the inverted U-shaped body **114** and a bar **234** extends to the rear R of the inverted U-shaped body **114**. The bars **232, 234** may have outwardly extending arcuate portions **236, 238** at the respective ends that compress and flatten to accommodate the thickness. Moreover the bars **232, 234** may be substantially equal to the length of the bars **124, 126, 128, 130**. In another embodiment, the bars **232, 234** may be greater than or less than the length of the bars **124, 126, 128, 130**.

(33) Two breakaway sections **140, 142** are defined along the respective two stems **118, 120** of the inverted U-shaped body **114**. Additionally, as shown, the notch **144** is formed between the bars **124, 126** and the notch **146** is formed between the bars **128, 130**. The open window **116** includes an upper edge **148**. The open span **150** is proximally interposed between the breakaway sections **140, 142** and the open span **150** forms a portion of the open window **116**. As shown, the crossbar **230** intersects the notch **144** forming subnotches **240, 242** between the bars **124, 126** and the crossbar **230**. Similarly, subnotches **244, 246** are formed between the bars **128, 130** and the crossbar **230**. It should be appreciated that although the I-shaped base **122** with the crossbar **230** is depicted with a particular U-shaped body **114**, it should be appreciated that the I-shaped base **122** with the crossbar **230** may be utilized with any of the leveling devices **10** presented herein, including the leveling devices **10** of FIG. 1A, FIG. 10, FIG. 13, and FIG. 14, for example.

(34) The order of execution or performance of the methods and techniques illustrated and described herein is not essential, unless otherwise specified. That is, elements of the methods and techniques may be performed in any order, unless otherwise specified, and that the methods may include more or less elements than those disclosed herein. For example, it is contemplated that executing or performing a particular element before, contemporaneously with, or after another element are all possible sequences of execution.

(35) While this invention has been described with reference to illustrative embodiments, this description is not intended to be construed in a limiting sense. Various modifications and combinations of the illustrative embodiments as well as other embodiments of the invention, will be apparent to persons skilled in the art upon reference to the description. It is, therefore, intended that the appended claims encompass any such modifications or embodiments.

Claims

1. A tile leveling device comprising: a body defining an open window; a base orthogonally coupled to the body, the base extending to a front of the body and the base extending to a rear of the body; a base to body coupling including a frangible breakaway section, the base and body being integral prior to frangible separation, the frangible breakaway section, upon breaking, frangibly separating the body from the base; a spacer integral with the body and extending from at least one of the front and rear thereof, the combination of the spacer and the body setting a predetermined joint distance that is greater than a thickness of the body for the tile leveling device within a single product; and a wedge device including a wedge member having a tapered surface configured to penetrate the open window and exert force against at least one tile.
2. The tile leveling device as recited in claim 1, wherein the wedge member further comprises teeth along the tapered surface, the teeth latch onto an upper edge of the open window.
3. The tile leveling device as recited in claim 1, further comprising proximate to the frangible

breakaway section, an open span forming a portion of the open window.

4. The tile leveling device as recited in claim 1, wherein the base further comprises an I-shaped base.

5. The tile leveling device as recited in claim 4, wherein the I-shaped base further comprises the I-shaped base having spaced first, second, third, and fourth bars extending transversely from the body, the first and second bars extending to the front of the body and the third and fourth bars extending to the rear of the body.

6. The tile leveling device as recited in claim 5, further comprising a first crossbar located between the first and third bars and a second crossbar located between the second and fourth bars.

7. A tile leveling device comprising: a body defining an open window; a base orthogonally coupled to the body, the base extending to a front of the body and the base extending to a rear of the body; and a spacer integral with the body and extending from at least one of the front and rear thereof, the combination of the spacer and the body setting a predetermined joint distance that is greater than a thickness of the body for the tile leveling device within a single product.

8. The tile leveling device as recited in claim 7, further comprising proximate to the frangible breakaway section, an open span forming a portion of the open window.

9. A tile leveling device comprising: a body defining an open window; a base orthogonally coupled to the body, the base extending to a front of the body and the base extending to a rear of the body; a base to body coupling including a frangible breakaway section, the base and body being integral prior to frangible separation, the frangible breakaway section, upon breaking, frangibly separating the body from the base; and a spacer integral with the body and extending from at least one of the front and rear thereof, the spacer setting a predetermined joint distance for the tile leveling device within a single product.

10. The tile leveling device as recited in claim 9, further comprising proximate to the frangible breakaway section, an open span forming a portion of the open window.
